The container method

May 28, 2026

Theorem 1 (Strong graph container lemma)**.** *For every $c > 0$, there exists $\delta > 0$ such that the following hold. Let G be an n-vertex graph of average degree d and maximum degree $\Delta(G) \leq cd$, and let $\tau = 2\delta/d$. There exists a collection $\mathcal{C}$ of subsets of vertices of G and a function $f : 2^{V(G)} \to \mathcal{C}$ such that:*

(i) every independent set I of G has a fingerprint *S such that $S \subseteq I \subseteq f(S)$ and $|S| \leq \tau n$,*

(ii) for every $C \in \mathcal{C}$, $|C| \leq (1-\delta)n$.

Theorem 2 (Strong hypergraph container lemma)**.** *For every $c > 0$, there exists $\delta > 0$ such that the following hold. Let $\mathcal{H}$ be a 3-uniform n-vertex hypergraph of average degree d and maximum degree $\Delta_1(\mathcal{H}) \leq cd$ and $\Delta_2(\mathcal{H}) \leq c\sqrt{d}$, and let $\tau = \delta/\sqrt{d}$. There exists a collection $\mathcal{C}$ of subsets of vertices of G and a function $f : 2^{V(G)} \to \mathcal{C}$ such that:*

(i) every independent set I of $\mathcal{H}$ has a fingerprint *S such that $S \subseteq I \subseteq f(S)$ and $|S| \leq \tau n$,*

(ii) for every $C \in \mathcal{C}$, $|C| \leq (1-\delta)n$.

1 Supersaturation of triangles

Exercise 1. Goodman bound: a combinatorial proof of supersaturation

1. Show that for every graph G, $\sum_{uv \in E(G)} \deg(u) + \deg(v) = \sum_{u \in V(G)} \deg(u)^2$
2. For every u and v, denote $\deg(u,v)$ the codegree of u, v, that is the number of common neighbours of u and v. Show that $\sum_{u \in V(G)} \deg(u)^2 \leq |E(G)||V(G)| + \sum_{uv \in E(G)} \deg(uv)$
3. Show that if G has $\gamma\binom{n}{2}$ edges, then G contains at least $\gamma(2\gamma-1)\frac{n^3}{6}$ triangles.

Exercise 2. A simple probabilistic proof

Let $\varepsilon > 0$ and and C be some large constant. Let G be a graph on $n \geq C$ vertices and at least $(\frac{1}{2}+\varepsilon)\binom{n}{2}$ edges.

1. Prove the reversed Markov inequality: let X be a random variable taking values in $[0,1]$,
$$\mathbb{P}(X \geq t) \geq \frac{\mathbb{E}[X] - t}{1-t} \text{ for every } t \in (0,1).$$
2. Let M be a random uniform subset of C vertices. Give a lower bound on the probability that $G[M]$ contains a triangle.
3. Let F be a random uniform triple of distinct vertices of G. Give a lower bound on the probability that $G[F]$ is a triangle.
4. Deduce that the number of triangles in G is $\Omega(n^3)$.

2 Other applications of the container method

Exercise 3.

We admit the following supersaturation result: for every $\varepsilon > 0$, there exists $\gamma > 0$ such that for every subset $A \subset [n]$ of size at least εn, A contains at least γn^2 triples $\{a, b, c\}$ such that $a - b = b - c$.

We denote $[n]_p$ the random subset of $[n]$ containing each element independently with probability p. The goal of this exercise is to prove the following theoem:

Theorem 3. *For every $\varepsilon > 0$, there exists $C > 0$ such that for every $p \geq C/\sqrt{n}$, every subset A of $[n]_p$ with size at least εpn contains an arithmetic progression of length 3.*

1. Prove the following container lemma for arithmetic progressions of length 3: For every γ, there exists $\delta > 0$ such that the following holds for every n. There exists a collection $\mathcal{C}$ of subsets of $[n]$ and a function $f : 2^n \to \mathcal{C}$ such that:
   (i) every subset $I \subset [n]$ without arithmetic progression of length 3 has a *fingerprint* S such that $S \subseteq I \subseteq f(S)$ and $|S| \leq 2\delta\sqrt{n}$,
   (ii) every $C \in \mathcal{C}$ contains at most γn^2 arithmetic progressions of length 3.
2. Prove Theorem 3.

Exercise 4.

The goal of this exercise is to prove the following theorem of Balogh and Solymosi, answering a question of Erdős:

Theorem 4. *There exists a set S of n points in the plane, containing no four points on a line, such that every subset of S of size $n^{5/6+o(1)}$ contains three points on a line.*

To do so, we consider $P = [m]^3$ the lattice in $\mathbb{R}^3$ and admit the following supersaturation theorem for collinear triples: for every $0 < \gamma < 1/2$ and every $S \subset [m]^3$ of size at least $m^{3-\gamma}$, there exist at least $m^{6-4\gamma-o(1)}$ collinear triples of points in S.

1. Prove the following container lemma: For every $m \in \mathbb{N}$, there exists a collection of containers $\mathcal{C}$ of $[m]^3$ such that
   (i) $|\mathcal{C}| \leq \exp(m^{5/3+o(1)})$
   (ii) $|C| \leq m^{8/3+o(1)}$ for every $C \in \mathcal{C}$
   (iii) every set of points S in general position (i.e. without three collinear points) is contained in some $C \in \mathcal{C}$
2. Let $p = m^{-1+o(1)}$ and consider R a random subset of points of $[m]^3$ containing each lattice point independently with probability p. Show that with high probability, R has size $m^{2+o(1)}$, R contains $o(|R|)$ collinear quadruples and for every $C \in \mathcal{C}$, $|R \cap C| \leq m^{5/3+o(1)}$.
3. Prove that there exists $A \subset R$ of size $m^{2+o(1)}$ with no collinear quadruple and containing no subset of $m^{5/3+o(1)}$ points in general position.
4. Finally, argue that A can be projected on a plane such that the projection A and $p(A)$ have the same size and the same number of collinear triples. Conclude.

★ Exercise 5.

Given a graph G, let $i(G)$ count the number of independent sets of G. The goal of this exercise is prove the following result of Alon: every d-regular graph on n vertices has $i(G) \leq 2^{(1+o(1))n/2}$ independent sets, where $o(1)$ is a function that tends to 0 as d tends to infinity.

1. Let $\beta > 0$. Prove that for every set of vertices A of size at least $\frac{n}{2} + \frac{\beta n^2}{2d}$, we have $e(A) \geq \beta\binom{n}{2}$.
2. Prove that every d-regular graph on n vertices has $i(G) \leq 2^{(1+o(1))n/2}$ independent sets.